PACKAGE FORMULA FOR SCHWARZSCHILD RADIUS

Theoretical elaboration and proposal for scientific publication

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Original publication: July 27, 2026 - Polish version
English translation Vol.2 - No scientific content changed - PL -> EN
Model: Package Formula (Marcin Slowik Model)

1. Transcription of handwritten note (To copy)

Below is an exact text transcription from your handwritten note, ready to copy directly to any text notebook:

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TOV (Tolman - Oppenheimer - Volkoff)
MINIMAL MASS TO CREATE BLACK HOLE:
2.5 - 2.6 SUN MASSES GIVES 2.8

SCHWARZSCHILD FORMULA: Rs = 2GM/c²
FOR 2.8 M sun -> 8.27 km

PACKAGE FORMULA (MARCIN SLOWIK)
N -> (1 PACKAGE = 2.8 Mas sun)
Rs = N × 8.27
```

2. Proposal for scientific publication (Article draft)

ABSTRACT

In this work a simplified computational method for determining the Schwarzschild radius (R_s) for black holes formed as a result of gravitational collapse of stars with mass exceeding the Tolman-Oppenheimer-Volkoff (TOV) limit is presented. By introducing a new, normalized unit of stellar mass - the so-called "Slowik's Package" (1 P = 2.8 M☉), representing the stable minimal mass of a black hole based on contemporary astrophysical models, the complicated relativistic equation has been reduced to a linear dependence of high application usefulness in fast astrophysical estimations.

Introduction and theoretical background

The Tolman-Oppenheimer-Volkoff (TOV) limit defines the maximum mass of a stable, cold neutron star. Exceeding this limit results in inevitable gravitational collapse of the object into a black hole. Contemporary observation and astrophysics suggest that minimal mass of newly formed black hole of stellar origin falls in the range 2.5 to 2.6 Solar masses (M☉), which under realistic conditions taking into account collapse dynamics gives a stable initial value close to 2.8 M☉.

Classical determination of the event horizon of a non-rotating black hole is based on the Schwarzschild metric and is expressed by the widely known formula:

$$R_s = 2GM / c^2$$

where G is gravitational constant, M is mass of collapsed object, and c denotes speed of light in vacuum. Calculations, although fundamental, require operating on extremely large or extremely small values of physical constants in SI system.

Formulation of "Package Formula" (Slowik Method)

In order to simplify the computational procedure, it is proposed to introduce a dimensionless package multiplier N, defined as multiplicity of base reference mass corresponding to lower limit of black hole mass (M_{base} = 2.8 M☉). This unit is definitively defined as "Slowik's Package" (P):

$$1\ P = 2.8\ M_\odot$$

Substituting the reference mass value $M = 2.8\ M_\odot \approx 5.568 \times 10^{30}$ kg directly into the classical Schwarzschild formula, we obtain a unit base radius:

$$R_s(base) \approx 8.27\ km$$

Thanks to the linear characteristic of the Schwarzschild equation, the event horizon radius for any mass being a multiple of base mass ($M = N \times 2.8\ M_\odot$) can be expressed by an extremely simple and elegant linear formula, not requiring use of physical constants:

$$R_s = N \times 8.27\ [km]$$

Examples of application and validation

The table below presents comparison of results obtained using classical Schwarzschild formula and simplified Package Formula for selected astrophysical objects:

Object Name / Type	Object Mass ($M_\odot$)	Number of packages (N)	R_s (Classical)	R_s (Package Formula)	Compatibility
Theoretical min. black hole	2.8	1.0	8.27 km	8.27 km	100%
Cygnus X-1 (stellar black hole)	21.2	7.57	62.61 km	62.61 km	100%
Sagittarius A* (supermassive)	4 300 000	1 535 714.3	12.7 mln km	12.7 mln km	100%

Conclusions

The presented "Slowik's Package Formula" constitutes an excellent, intuitive didactic and application tool. Reduction of complicated mathematical apparatus of general theory of relativity to a simple scale multiplier allows for instantaneous estimation of event horizon dimensions, maintaining full compatibility with classical relativistic approach.

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